Multi-View Clustering With Hybrid-Order Similarity Learning

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***Abstract*—Multi-view clustering (MVC) has attracted significant attention in recent years due to its ability to leverage heterogeneous features from multiple views. However, existing methods often lack the ability to jointly model first-order and topological relationships, which is crucial for capturing a more comprehensive clustering structure. In this article, we propose a novel multi-view clustering method with hybrid-order similarity learning (MCHL), which integrates multiple view-specific graphs while considering their first-order and topological correlations, and iteratively learns the view weights and the consensus graph within a unified framework. In addition, we impose a connectivity constraint on the consensus graph to ensure that data points belonging to the same cluster are properly connected within the same component. Extensive experiments on multiple benchmark datasets demonstrate the superior clustering performance of MCHL over the state-of-the-art methods.**

***Index Terms*—Data clustering, hybrid-order similarity learning, manifold learning, multi-view clustering (MVC).**

Nomenclature

$X^{(v)} \in \mathbb{R}^{d_v \times n}$	vth view of a multi-view dataset.
n	Number of samples in the dataset.
m	Number of views in the dataset.
c	Number of clusters.
d_v	Dimension of the vth view in the dataset.
$G^{(v)} \in \mathbb{R}^{n \times n}$	Input similarity matrix for the vth view.
$D^{(v)} \in \mathbb{R}^{n \times n}$	Degree matrix of $G^{(v)}$.
$Z^{(v)} \in \mathbb{R}^{n \times n}$	Learned similarity matrix for the vth view.
$S \in \mathbb{R}^{n \times n}$	Unified similarity matrix.
$F \in \mathbb{R}^{n \times c}$	Indicator matrix of data.
$L_S \in \mathbb{R}^{n \times n}$	Laplacian matrix of S.
$\phi^{(v)}$	Weight coefficient for the vth view.

I. Introduction

IN MANY real-world applications, data can be collected from different sources or described by different types of features, leading to multiple views of the same underlying structure. For example, an image can be depicted by different feature descriptors. A news event may be reported by different media in various languages. Such data are referred to as multi-view data, where each data entity is described by multiple distinct sets of features. Multi-view data provide complementary information from diverse perspectives, which can substantially improve various data mining tasks. As a result, multi-view clustering (MVC) has recently gained considerable attention due to its advantage in fusing common and complementary information from multiple views to enhance clustering performance [1], [2], [3].

In comparison with traditional single-view clustering methods, MVC can significantly improve clustering performance by leveraging consistent and complementary information across multiple views. Existing MVC methods can be broadly classified into several categories. One popular category is based on nonnegative matrix factorization (NMF), which has been widely applied to multi-view data processing [1], [4], [5], [6], [7], [8]. Another line of research focuses on learning a consensus latent subspace representation shared across views [2], [9], [10], [11], [12], [13], including subspace learning and NMF-based techniques. Tensor learning-based MVC methods capture complex relationships across different views by leveraging multidimensional data representations [14], [15], [16], [17], [18]. Another important category involves fusing graph structures from multiple views and performing graph partitioning on the fused graph to obtain clustering results [19], [20], [21], [22], [23], [24]. Recently, deep learning-based MVC methods, particularly those utilizing neural networks, have gained popularity due to their ability to automatically learn complex feature representations [25], [26], [27], [28], [29], [30], [31], [32].

Among the existing MVC techniques, graph-based MVC is one of the most representative paradigms. Graphs are fundamental data structures for representing relationships among distinct entities, where nodes represent entities and edges indicate relationships or similarities. A key step in graph-based MVC is constructing a similarity graph, which directly impacts clustering quality. Different methods adopt various strategies to learn similarity graphs. Some approaches construct similarity graphs based on manifold similarity learning [19], while others leverage subspace information for similarity learning [33], [34], [35]. These strategies are widely used but share a common limitation: they primarily rely on first-order connections (direct associations) in the original data graph. Although first-order connections effectively capture local structures, they fail to capture broader structural relationships in the data.

Currently, research on constructing similarity graphs that incorporate higher-order connections is still limited. Some recent studies attempt to learn similarity graphs using topological structures [36], [37], [38]. The core idea is to utilize second-order associations, which measure relationships

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between data points based on shared neighbors. While these methods enhance the representation of topological structures, they often fail to preserve local structures. Therefore, designing a similarity graph that simultaneously captures both first-order local structures and second-order topological structures remains a challenging problem.

In light of this, this article presents a novel method termed multi-view clustering with hybrid-order similarity learning (MCHL). It first performs similarity learning based on neighborhood structures, which capture the local relationships between data points and thus encourage nearby points to be grouped together. MCHL calculates pairwise similarity based on the direct neighbors of data points to preserve the local geometric structures within each view. Next, while preserving local geometric structures, it also captures broader global structures, referred to here as topological structures, by considering the connectivity patterns of data points. Here, second-order similarities are computed between data points through their shared neighbors to preserve the global topological structure of the data. To integrate both neighborhood and topological structures, a unified optimization model is designed to integrate both types of similarity learning. In particular, it constructs a unified similarity graph by simultaneously exploiting neighborhood (first-order) and topological (second-order) structures in each view. The topological correlation captures structural similarity induced by shared neighborhood connectivity patterns. Although related to higher-order structural modeling, the proposed formulation explicitly restricts the interaction to second-order relationships, which enables effective characterization of global structures while avoiding the potential instability and noise accumulation often introduced by higher-order graph modeling. This hybrid-order learning strategy enables the final clustering results to be derived from a comprehensive data representation that preserves both local relationships and global structural consistency.

For clarity, the main contributions of this article are summarized below.

1) The first-order neighborhood similarity and second-order topological similarity are jointly modeled into a hybrid-order similarity learning framework, which can simultaneously capture local geometric relationships and global structural dependencies in data.
2) By integrating neighborhood structures with topological structures, a more informative similarity graph is learned to better reflect the intrinsic structure of multi-view data.
3) The proposed method directly derives the final clustering results from the learned unified similarity graph, eliminating the need for additional postprocessing steps and enabling clustering to be achieved within a unified optimization model.

The rest of this article is organized as follows. In Section II, we review the related works, especially in multi-view graph learning. Section III presents the proposed graph-based MVC framework. In Section IV, we discuss the optimization problem of this framework and present an efficient algorithm to solve it. Extensive experimental results are reported in Section V. Finally, we conclude this article in Section VI.

II. Related Work

A. Multi-View Graph-Based Clustering

Multi-view graph-based clustering enhances clustering performance by fusing graph structures from multiple views and leveraging complementary information across views [19], [20], [21], [22], [23], [24], [39].

For instance, Wang et al. [19] jointly learned and fused multiple view graphs into a unified graph matrix, automatically weighted each view, and directly produced the final clusters while imposing a rank constraint on the Laplacian matrix without requiring any tuning parameters. Huang et al. [20] simultaneously and explicitly leveraged multi-view consistency and diversity, and fused the consistent parts into a target graph with a clear clustering structure. Huang et al. [21] automatically learned optimal weights for each bipartite graph without introducing additional parameters, and efficiently integrated these graphs to obtain an optimal consensus graph. Wang et al. [22] integrated a multigraph autoencoder (M-GAE) with a multi-view mutual information maximization module (MMIM) and a graph fusion network (GFN), enabling the joint learning of a common representation that captures complementary information from multiple views while preserving neighborhood similarity within each view. Li et al. [23] proposed a scalable and parameter-free graph fusion framework for MVC, which jointly learns a compatible graph across multiple views in a self-supervised weighting manner, eliminates weight-related hyperparameters, and incorporates a connectivity constraint to directly indicate clusters. Liang et al. [24] simultaneously and explicitly modeled both multi-view consistency and inconsistency in a unified objective function, which enables the iterative learning of consistent and inconsistent parts of each single-view graph, as well as the unified graph that fuses the consistent parts. Kang et al. [39] fused different views into one graph by approximating the original graph of each view while maintaining an explicit cluster structure.

B. Topological Structure

In real-world applications, high-dimensional data are ubiquitous, and high-dimensional vector spaces are often too sparse to be processed effectively [40]. To map high-dimensional data to lower-dimensional spaces, nonlinear manifold learning [41], [42], [43], [44], [45] has gained widespread attention. These methods primarily focus on learning the manifold structure embedded in high-dimensional spaces.

In [37], topological relationships are exploited to reveal latent group structures and measure the collectiveness of samples. Two samples may have low direct similarity in the original space but high topological similarity if they are connected through a sequence of neighboring samples.

Given a similarity graph G, if two samples are highly consistent, it is assumed that their topological correlations with other points should be similar. To compute the topological similarity between samples, Wang et al. [37] proposed the

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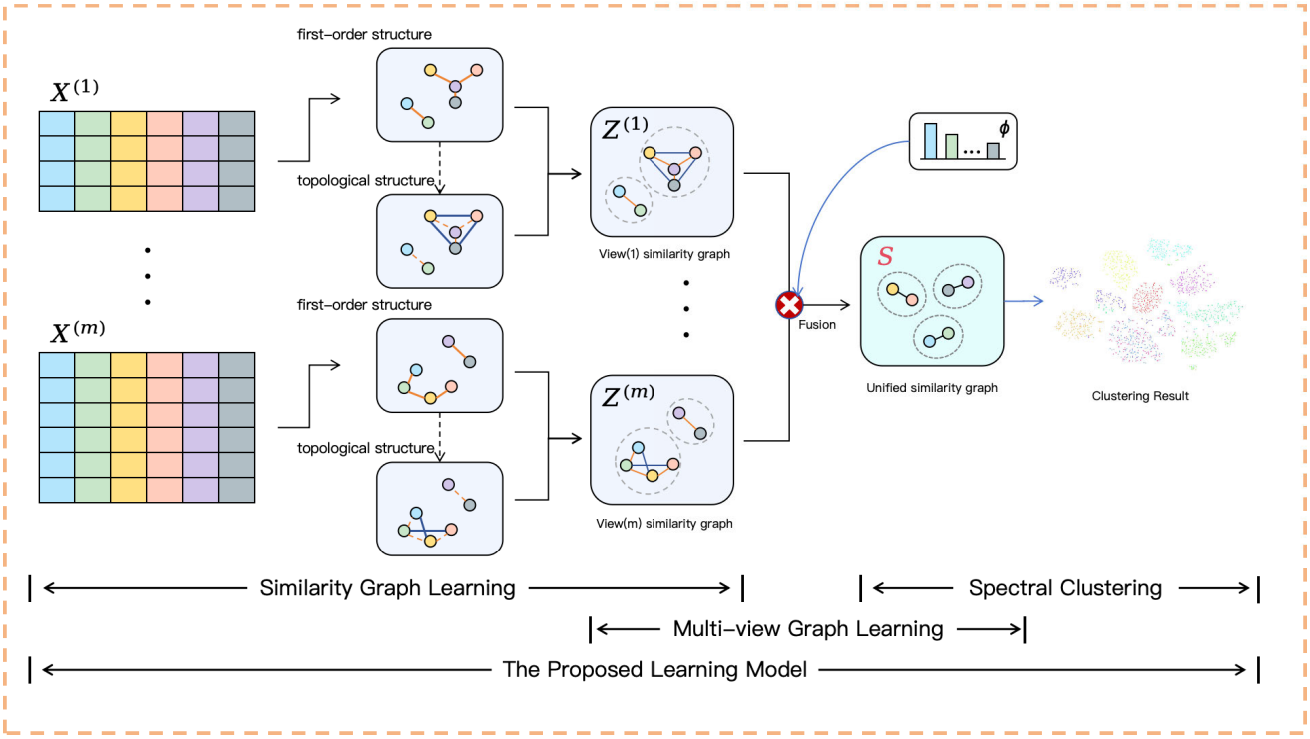


Fig. 1. Proposed framework of MCHL. The multi-view data $\{X^{(v)}\}_{v=1}^{m} \in \mathbb{R}^{d_v \times n} (1 \le v \le m)$ are first exploited to learn the first-order neighborhood similarity, followed by the extraction of topological structure similarity. These two similarities are integrated to construct a similarity graph $\{Z^{(v)}\}_{v=1}^{m} \in \mathbb{R}^{n \times n}$ for each view. Subsequently, the view-specific similarity graphs are fused into a unified consensus graph $S \in \mathbb{R}^{n \times n}$, from which the final clustering results are directly obtained.

following formula:

$$\min_{\{\mathbf{Z}\}} \frac{1}{2} \sum_{i,j,k=1}^{n} G_{jk} \left(Z_{ij} - Z_{ik}\right)^2 + \xi \|\mathbf{Z} - \mathbf{I}\|_F^2 \tag{1}$$

where Z_{ij} represents the topological correlation between sample i and sample j. The first term of the formula applies a smoothing constraint to satisfy the above assumption, ensuring that if samples j and k exhibit high consistency, their topological correlations with sample i are similar. The second term is a fitting constraint to prevent trivial solutions for Z, and the parameter ξ balances these two terms.

III. Proposed Framework

In this section, we present the proposed **MCHL** approach (as shown in Fig. 1) in detail. Specifically, the notation used throughout this article is summarized in Nomenclature. Section III-A introduces first-order neighborhood structure learning, Section III-B presents topological structure learning, and Section III-C describes multi-view graph fusion and the complete objective function.

A. First-Order Neighborhood Structure Learning

In the context of MVC, the data are often represented in multiple views (feature spaces), each providing different perspectives on the same set of objects. The key challenge is to effectively integrate information from all views to produce a robust clustering solution that captures the underlying cluster structure of the data. First-order neighborhood learning is focused on capturing the direct neighbor relationships within each view, which are critical for understanding the fine-grained structure of the data.

Consider a dataset $X = \{X_1, X_2, \ldots, X_n\}$ containing n samples, where X_i represents the ith sample of the dataset. In the case of a multi-view dataset, each sample can be described by different feature sets across multiple views. Accordingly, the multi-view dataset can be expressed as $\{X^{(v)}\}_{v=1}^{m}$, where $X^{(v)} \in \mathbb{R}^{d_v \times n}$ is the data matrix of the vth view, m denotes the number of views, and d_v is the dimension of the vth view.

The goal of MVC is to partition n data points into c clusters such that:

1) Data points within the same cluster are more similar to each other than to those in different clusters.
2) Information from multiple views is effectively integrated to reflect both consistent and view-specific characteristics.

Mathematically, this can be viewed as finding a consistent assignment matrix $Q \in \{0,1\}^{n \times c}$ while learning a similarity matrix $Z^{(v)}$ for each view. Wang et al. [19] suggest that a larger Euclidean distance between two data points corresponds to a smaller similarity value, and a smaller Euclidean distance corresponds to a larger similarity value. The first-order similarity can be mathematically modeled as follows:

$$\min_{\{\mathbf{Z}^{(v)}\}} \sum_{v=1}^{m} \sum_{i,j=1}^{n} \left\| X_i^{(v)} - X_j^{(v)} \right\|_2^2 Z_{ij}^{(v)}$$
$$\text{s.t.} \quad \forall v,\ Z_{ii}^{(v)} = 0,\ Z_{ij}^{(v)} \ge 0. \tag{2}$$

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The study in [46] demonstrates that sparse representation is highly effective in handling data loss and outliers in high-dimensional datasets. Incorporating the normalization constraint $\mathbf{1}^T z_i^{(v)} = 1$, where $z_i^{(v)} \in \mathbb{R}^{n\times 1}$ is a vector whose j-th element is $Z_{ij}^{(v)}$, problem (2) becomes

$$\begin{aligned}\min_{\{\mathbf{Z}^{(v)}\}} &\sum_{v=1}^{m}\sum_{i,j=1}^{n}\left\|X_i^{(v)} - X_j^{(v)}\right\|_2^2 Z_{ij}^{(v)} \\ \text{s.t.} \quad &\forall v,\ Z_{ii}^{(v)} = 0, Z_{ij}^{(v)} \geq 0,\ \mathbf{1}^T z_i^{(v)} = 1. \end{aligned} \tag{3}$$

B. Topological Structure Learning

Two samples may have low direct similarity; however, if they are connected through a sequence of common neighbors, their topological similarity can be high [37]. Therefore, topological relationships provide an additional measure of sample consistency. In this section, we introduce a topological structure learning method to reveal latent cluster structure by analyzing such relationships among samples.

Before exploring the topological relationship between two samples, we first define the input similarity graph. Let $G^{(1)}, G^{(2)}, \ldots, G^{(m)}$ be the input similarity graphs for the m views, with $G^{(v)} \in \mathbb{R}^{n\times n}(1 \leq v \leq m)$. While the element $G_{ij}^{(v)}$ represents the input similarity between the two samples, we can capture the topological relationship between samples by solving

$$\begin{aligned}\min_{\{\mathbf{Z}\}} &\sum_{i,j,k=1}^{n} G_{j,k}\left(Z_{ij} - Z_{ik}\right)^2 \\ \text{s.t.} \quad &\left(z_i\right)^T \mathbf{1} = 1,\ Z_{ij} \geq 0 \end{aligned} \tag{4}$$

where we constrain the sum of each row of Z to be 1, and require every element of Z to be nonnegative. For the MVC task, we extend the formulation to its multi-view version, which is represented as

$$\begin{aligned}\min_{\{\mathbf{Z}^{(v)}\}} &\sum_{v=1}^{m}\sum_{i,j,k=1}^{n} G_{j,k}^{(v)}\left(Z_{ij}^{(v)} - Z_{ik}^{(v)}\right)^2 \\ \text{s.t.} \quad &\left(z_i^{(v)}\right)^T \mathbf{1} = 1,\ Z_{ij}^{(v)} \geq 0. \end{aligned} \tag{5}$$

According to (5), a sample with many strongly connected neighbors may exert a disproportionately large influence on the local structure. To address this imbalance, we normalize the topological similarity Z_{ij} to balance the influence of each sample by node degree, leading to

$$\begin{aligned}\min_{\{\mathbf{Z}^{(v)}\}} &\sum_{v=1}^{m}\sum_{i,j,k=1}^{n} G_{j,k}^{(v)}\left(\frac{Z_{ij}^{(v)}}{\sqrt{D_{jj}^{(v)}}} - \frac{Z_{ik}^{(v)}}{\sqrt{D_{kk}^{(v)}}}\right)^2 \\ \text{s.t.} \quad &\left(z_i^{(v)}\right)^T \mathbf{1} = 1,\ Z_{ij}^{(v)} \geq 0 \end{aligned} \tag{6}$$

where $D^{(v)}$ is the degree matrix of $G^{(v)}$.

C. Multi-View Graph Fusion

In similarity-based clustering, the scaling of the similarity matrix does not alter its functional role or significance. Specifically, for any nonzero scalar k, the similarity matrix Z is equivalent to $k \cdot Z$.

Under the assumption that scaling does not influence the results, we need to scale the similarity matrices and then merge them into a unified similarity matrix. Once we have learned the clustering structure across multiple views, we can obtain consolidated similarity information by integrating the similarity graphs Z from the various views. The unified similarity matrix S can be learned as follows:

$$\begin{aligned}\min_{\{S,\phi^{(v)}\}} &\sum_{v=1}^{m}\left|\left|S - \phi^{(v)}Z^{(v)}\right|\right|_F^2 \\ \text{s.t.}\ &s_i^T\mathbf{1} = 1,\ S \geq 0,\ \phi^{(v)} \geq 0,\ \sum_{v=1}^{m}\phi^{(v)} = 1. \end{aligned} \tag{7}$$

This encourages the weighted view-specific similarity graphs to agree with the unified similarity matrix S, where $\phi^{(v)}$ represents a learnable coefficient. To ensure the uniqueness of the scaling results, we impose the constraint that the sum of $\phi^{(v)}$ is equal to 1. At this stage, we have obtained the first-order neighborhood structure of multi-view data through (3), learned the topological structure via (6), and derived the unified similarity matrix through (7). Consequently, by integrating (3), (6), and (7), we can formulate the following expression:

$$\begin{aligned}\min_{\{Z^{(v)},S,\phi^{(v)}\}} &\frac{1}{2}\sum_{v=1}^{m}\sum_{i,j,k=1}^{n} G_{j,k}^{(v)}\left(\frac{Z_{ij}^{(v)}}{\sqrt{D_{jj}^{(v)}}} - \frac{Z_{ik}^{(v)}}{\sqrt{D_{kk}^{(v)}}}\right)^2 \\ &+ \sum_{v=1}^{m}\sum_{i,j=1}^{n}\left\|X_i^{(v)} - X_j^{(v)}\right\|_2^2 Z_{ij}^{(v)} + \sum_{v=1}^{m}\left\|Z^{(v)} - I\right\|_F^2 \\ &+ \sum_{v=1}^{m}\left|\left|S - \phi^{(v)}Z^{(v)}\right|\right|_F^2 \\ \text{s.t.}\quad &s_i^T\mathbf{1} = 1,\ S_{ij} \geq 0,\ S \geq 0,\ \left(z_i^{(v)}\right)^T\mathbf{1} = 1 \\ &Z_{ij}^{(v)} \geq 0,\ \phi^{(v)} \geq 0,\ \sum_{v=1}^{m}\phi^{(v)} = 1. \end{aligned} \tag{8}$$

The above formulation does not directly yield clustering results from the unified similarity matrix S, necessitating additional clustering algorithms or steps. To address this issue, we introduce a Laplacian rank constraint.

Theorem 1: The multiplicity r of the eigenvalue 0 of the Laplacian matrix L_S is equal to the number of connected components in the graph of the unified matrix S.

The proof of this theorem can be found in [47] and [48]. The objective of our proposed method is to identify a consensus graph that exhibits an explicit clustering structure. Ideally, if the dataset contains c clusters, we aim to find a graph with c connected components. This can be expressed as $rank(L_S) = n - c$, where L_S represents the Laplacian matrix of the unified graph S. While we can incorporate a rank constraint in (8), it is difficult to optimize directly due to the nonlinear nature of $rank(L_S)$.

Let σ_i represent the ith smallest eigenvalue of L_S. The rank constraint $rank(L_S) = n - c$, with $c = r$, is equivalent to

$$\sum_{i=1}^{c}\sigma_i\left(L_S\right) = 0 \tag{9}$$

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Since L_S is positive semidefinite, we have $\sigma_i \geq 0$ [47]. By applying Ky Fan's Theorem [49], we can reformulate the problem into a more manageable form as follows:

$$\sum_{i=1}^{c} \sigma_i (L_S) = \min_{F^T F = I} \mathrm{Tr}\left(F^T L_S F\right) \tag{10}$$

where F denotes the cluster indicator matrix. By incorporating (10) into (8), we obtain the final objective function as follows:

$$\begin{aligned} \min_{\{Z^{(v)},S,\phi^{(v)},F\}} \quad & \frac{1}{2}\sum_{v=1}^{m}\sum_{i,j,k=1}^{n} G_{j,k}^{(v)} \left(\frac{Z_{ij}^{(v)}}{\sqrt{D_{jj}^{(v)}}} - \frac{Z_{ik}^{(v)}}{\sqrt{D_{kk}^{(v)}}}\right)^2 \\ & + \alpha \sum_{v=1}^{m}\sum_{i,j=1}^{n} \left\| X_i^{(v)} - X_j^{(v)} \right\|_2^2 Z_{ij}^{(v)} \\ & + \beta \sum_{v=1}^{m} \left\| Z^{(v)} - I \right\|_F^2 + \gamma \sum_{v=1}^{m} \left|\left| S - \phi^{(v)} Z^{(v)} \right|\right|_F^2 \\ & + \lambda \, Tr\left(F^T L_S F\right) \\ \text{s.t.} \quad & s_i^T \mathbf{1} = 1, \; S \geq 0, \; \left(z_i^{(v)}\right)^T \mathbf{1} = 1 \\ & Z_{ij}^{(v)} \geq 0, \; \phi^{(v)} \geq 0, \; \sum_{v=1}^{m} \phi^{(v)} = 1, \; F^T F = I. \end{aligned} \tag{11}$$

The objective function in (11) is composed of several terms with distinct roles. The *first term* preserves the second-order structural information by modeling topological similarity through second-order neighborhood consistency. The *second term* encodes the first-order similarity to maintain local pairwise relationships. The *third term* serves as a fitting constraint to prevent trivial solutions of the similarity matrix $Z^{(v)}$. The *fourth term* enforces consistency between view-specific similarity matrices and the consensus graph S via adaptive view weights $\phi^{(v)}$. Finally, the *Laplacian regularization term* with coefficient λ constrains the rank of L_S, ensuring exactly c connected components corresponding to the target clusters.

The model introduces three key hyperparameters: α governs the weight of first-order similarity, β modulates the impact of the fitting constraint, and γ controls the intensity of the fusion term.

IV. Optimization

A. Update F

With the variables $\phi^{(v)}$, S, and $Z^{(v)}$ fixed, F can be optimized according to the following problem:

$$\begin{aligned} \min_{\{F\}} \quad & Tr\left(F^T L_S F\right) \\ \text{s.t.} \quad & F^T F = I. \end{aligned} \tag{12}$$

This is a common spectral clustering problem, and its optimal solution F can be obtained by calculating the eigenvectors corresponding to the c smallest eigenvalues of L_S.

B. Update $Z^{(v)}$

When other variables $\phi^{(v)}$, S, and F are fixed, the subproblem for $Z^{(v)}$ becomes

$$\begin{aligned} \min_{\{Z^{(v)}\}} \quad & \frac{1}{2}\sum_{v=1}^{m}\sum_{i,j,k=1}^{n} G_{j,k}^{(v)} \left(\frac{Z_{ij}^{(v)}}{\sqrt{D_{jj}^{(v)}}} - \frac{Z_{ik}^{(v)}}{\sqrt{D_{kk}^{(v)}}}\right)^2 \\ & + \alpha \sum_{v=1}^{m}\sum_{i,j=1}^{n} \left\| X_i^{(v)} - X_j^{(v)} \right\|_2^2 Z_{ij}^{(v)} + \beta \sum_{v=1}^{m} \left\| Z^{(v)} - I \right\|_F^2 \\ & + \gamma \sum_{v=1}^{m} \left|\left| S - \phi^{(v)} Z^{(v)} \right|\right|_F^2 \\ \text{s.t.} \quad & \left(z_i^{(v)}\right)^T \mathbf{1} = 1, \; Z_{ij}^{(v)} \geq 0. \end{aligned} \tag{13}$$

Note that problem (13) is independent for different v, so we can solve the following problem separately for each v:

$$\begin{aligned} \min_{\{Z^{(v)}\}} \quad & \frac{1}{2}\sum_{i,j,k=1}^{n} G_{j,k}^{(v)} \left(\frac{Z_{ij}^{(v)}}{\sqrt{D_{jj}^{(v)}}} - \frac{Z_{ik}^{(v)}}{\sqrt{D_{kk}^{(v)}}}\right)^2 \\ & + \alpha \sum_{i,j=1}^{n} \left\| X_i^{(v)} - X_j^{(v)} \right\|_2^2 Z_{ij}^{(v)} + \beta \left\| Z^{(v)} - I \right\|_F^2 \\ & + \gamma \left\| S - \phi^{(v)} Z^{(v)} \right\|_F^2 \\ \text{s.t.} \quad & \left(z_i^{(v)}\right)^T \mathbf{1} = 1, \; Z_{ij}^{(v)} \geq 0. \end{aligned} \tag{14}$$

For simplicity, D and G denote $D^{(v)}$ and $G^{(v)}$, respectively, in the following derivation. Let $d_{ij} = \|X_i^{(v)} - X_j^{(v)}\|_2^2$. Since for different i, (14) is independent, we can solve the following subproblem separately for each i:

$$\begin{aligned} \min_{\left\{z_i^{(v)}\right\}} \quad & \left(z_i^{(v)}\right)^T z_i^{(v)} - \left(z_i^{(v)}\right)^T D^{-\frac{1}{2}} G D^{-\frac{1}{2}} z_i^{(v)} \\ & + \alpha \sum_{j=1}^{n} d_{ij} Z_{ij}^{(v)} + \beta \left\| z_i^{(v)} - e_i \right\|_2^2 \\ & + \gamma \left\| s_i - \phi^{(v)} z_i^{(v)} \right\|_2^2 \\ \text{s.t.} \quad & \left(z_i^{(v)}\right)^T \mathbf{1} = 1, \; Z_{ij}^{(v)} \geq 0 \end{aligned} \tag{15}$$

where $\mathbf{z}_i^{(v)}$ is a vector with its jth element being $Z_{ij}^{(v)}$ of $Z^{(v)}$ and e_i denotes the ith standard basis vector corresponding to the ith row of the identity matrix I. Let $\mathbf{d}_i$ be a vector whose jth element is d_{ij}, and denote $\mathbf{A} = (1 + \beta + \gamma(\phi^{(v)})^2)\mathbf{I} - D^{-1/2} G D^{-1/2}$ and $\mathbf{b} = \alpha \mathbf{d}_i - 2\beta \mathbf{e}_i - 2\gamma \boldsymbol{\phi}^{(v)} \mathbf{s}_i$, (15) can be transformed to

$$\begin{aligned} \min_{\left\{\mathbf{z}_i^{(v)}\right\}} \quad & \left(\mathbf{z}_i^{(v)}\right)^T \mathbf{A} \mathbf{z}_i^{(v)} + \mathbf{b}^T \mathbf{z}_i^{(v)} \\ \text{s.t.} \quad & \left(\mathbf{z}_i^{(v)}\right)^T \mathbf{1} = 1, \; Z_{ij}^{(v)} \geq 0. \end{aligned} \tag{16}$$

It is a convex quadratic optimization problem, which can be solved using the augmented Lagrangian multiplier (ALM) method. Thus, the augmented Lagrangian function for (16) can be transformed as follows using ALM:

$$\min_{\left\{\mathbf{z}_i^{(v)}, p\right\}} \quad \left(\mathbf{z}_i^{(v)}\right)^T \mathbf{A} p + \mathbf{b}^T \mathbf{z}_i^{(v)}$$

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$$+\frac{\delta}{2}\left\|\mathbf{z}_i^{(v)}-p+\frac{1}{\delta}q\right\|_2^2$$
$$\text{s.t.}\quad \left(\mathbf{z}_i^{(v)}\right)^T\mathbf{1}=1, Z_{ij}^{(v)}\geq 0 \tag{17}$$

where δ and q are the corresponding penalty coefficient and parameter, respectively. The third term in (17) is a penalty function, which ensures that $p=\mathbf{z}_i^{(v)}$.

Based on (17), the problem in (16) can be solved through iterative optimization.

1) Update p: When $\mathbf{z}_i^{(v)}$ is fixed, we can get the Lagrange function of (17) w.r.t. p. By setting its derivative to zero, we can obtain the solution for p as

$$p=\mathbf{z}_i^{(v)}-\frac{1}{\delta}\left(\mathbf{A}^T\mathbf{z}_i^{(v)}-q\right). \tag{18}$$

2) Update $\mathbf{z}_i^{(v)}$: When p is fixed, we can get the Lagrange function of (17) w.r.t. $\mathbf{z}_i^{(v)}$ as

$$\min_{\left\{\mathbf{z}_i^{(v)}\right\}}\quad \left\|\mathbf{z}_i^{(v)}-p+\frac{\mathbf{A}p+\mathbf{b}+q}{\delta}\right\|_2^2$$
$$\text{s.t.}\quad \left(\mathbf{z}_i^{(v)}\right)^T\mathbf{1}=1, Z_{ij}^{(v)}\geq 0. \tag{19}$$

A closed-form solution can be obtained through the optimization algorithm in [50].

According to the ALM principle, δ is increased at each iteration, and q is updated as $q=q+\delta(\mathbf{z}_i^{(v)}-p)$.

C. Update S

By fixing all variables except S, the subproblem for S becomes

$$\min_{\{S\}}\quad \sum_{v=1}^{m}\gamma\left\|S-\phi^{(v)}Z^{(v)}\right\|_F^2+2\lambda\, Tr\left(F^T L_S F\right)$$
$$\text{s.t.}\quad \mathbf{s}_i^T\mathbf{1}=1,\ S_{ij}\geq 0. \tag{20}$$

By setting $\Theta_{ij}=\|f_i-f_j\|_2^2$, we can rewrite (20) as

$$\min_{\{S\}}\quad \sum_{i,j=1}^{n}\lambda\Theta_{ij}S_{ij}+\gamma\sum_{v=1}^{m}\sum_{i,j=1}^{n}\left(S_{ij}-\phi^{(v)}Z_{ij}^{(v)}\right)^2$$
$$\text{s.t.}\quad \mathbf{s}_i^T\mathbf{1}=1,\ S_{ij}\geq 0. \tag{21}$$

Note that the problem is independent for each i. Let $\tilde{\mathbf{z}}=\sum_{v=1}^{m}\phi^{(v)}Z^{(v)}$, then we can solve the problem for each i by

$$\min_{\{S\}}\quad \gamma\left\|\mathbf{s}_i-\left(\tilde{\mathbf{z}}_i-\frac{\lambda}{2\gamma}\theta_i\right)\right\|_F^2$$
$$\text{s.t.}\quad \mathbf{s}_i^T\mathbf{1}=1,\ S_{ij}\geq 0 \tag{22}$$

where $\tilde{\mathbf{z}}_i$ and θ_i are two column vectors corresponding to the ith rows of $\tilde{\mathbf{z}}$ and Θ, respectively. Equation (22) can be solved independently for each row of S. Let $\omega=s_i$ and $\psi=\tilde{\mathbf{z}}_i-\lambda/2\gamma\theta_i$, then the optimization problem is equivalent to

$$\min_{\{\omega\}}\quad \|\omega-\psi\|_2^2-\nu\left(\omega^T\mathbf{1}-1\right)-\tau^T\omega \tag{23}$$

where ν is a scalar and τ is a Lagrangian coefficient vector. Assume that the optimal solution to the proximal problem (23) is ω^*, with the associated Lagrangian coefficients being ν^* and τ^*. According to the Karush–Kuhn–Tucker (KKT) conditions described in [51], we can obtain the following equation:

$$\begin{cases}\forall i;\quad \omega_i^*-\psi_i-\nu^*-\tau_i^*=0\\ \forall i;\quad \omega_i^*\geq 0\\ \forall i;\quad \tau_i^*\geq 0\\ \forall i;\quad \omega_i^*\tau_i^*=0\end{cases} \tag{24}$$

where ω_i^* is the ith scalar element of ω^*. According to the first term of (24) and constraint $\mathbf{1}^T\omega^*=1$, thus we have $\nu^*=(1-\mathbf{1}\psi-\mathbf{1}\tau)/n$, then $\omega^*=(\psi-\mathbf{1}\mathbf{1}^T/n\psi+1/n\mathbf{1}-\mathbf{1}^T\tau^*/n\mathbf{1})+\tau^*$.

Setting $\tilde{\tau}^*=\mathbf{1}\tau^*/n$ and $\mathbf{u}=\psi-\mathbf{1}\mathbf{1}^T/n\psi+1/n\mathbf{1}$, then we can obtain $\omega^*=\mathbf{u}+\tau^*-\tilde{\tau}^*\mathbf{1}$. Thus, for all i, we have

$$\omega_i^*=u_i+\tau_i^*-\tilde{\tau}^*. \tag{25}$$

Based on (24) and (25), we have $u_i+\tau^*-\tilde{\tau}^*=(u_i-\tilde{\tau}^*)_+$, where $x_+=max(x,0)$. Then we have

$$\omega_i^*=(u_i-\tilde{\tau}^*)_+. \tag{26}$$

It is evident that if we know τ, we can easily derive the optimal solution ω.

In order to obtain τ, we rewrite (25) as $\tau_i^*=\omega_i^*+\tilde{\tau}^*-u_i$. Similarly, based on (24), we can derive $\tau_i^*=(\tilde{\tau}^*-u_i)_+$. Substituting it into $\tilde{\tau}^*$ yields $\tilde{\tau}^*=1/n\sum_{i=1}^{n}(\tilde{\tau}^*-u_i)_+$. We can then define a function as follows:

$$f(\tilde{\tau})=\frac{1}{n}\sum_{i=1}^{n}(\tilde{\tau}^*-u_i)_+-\tilde{\tau}. \tag{27}$$

Setting $f(\tilde{\tau}^*)=0$, we can solve the root-finding problem to find $\tilde{\tau}^*$.

Noting that $\tau^*\geq 0$, $f'(\tilde{\tau})\leq 0$, and that $f'(\tilde{\tau})$ is a piecewise linear convex function, we can effectively find the root of $f(\tilde{\tau})=0$ using Newton's method, specifically

$$\tilde{\tau}_{t+1}=\tilde{\tau}_t-\frac{f(\tilde{\tau})}{f'(\tilde{\tau})}. \tag{28}$$

D. Update $\phi^{(v)}$

When other variables are fixed, updating the view weight can be derived from (7) and formulated as

$$\min_{\{\phi^{(v)}\}}\quad \sum_{v=1}^{m}\|S-\phi^{(v)}Z^{(v)}\|_F^2$$
$$\text{s.t.}\quad \sum_{v=1}^{m}\phi^{(v)}=1,\quad \phi^{(v)}\geq 0 \tag{29}$$

For each view, the Lagrange function of Equation (29) is

$$\mathcal{L}(\phi,\eta)=\sum_{v=1}^{m}\|S-\phi^{(v)}Z^{(v)}\|_F^2+\eta\left(\sum_{v=1}^{m}\phi^{(v)}-1\right), \tag{30}$$

where η is the Lagrange multiplier associated with the equality constraint.

Setting the derivative of $\mathcal{L}$ w.r.t. $\phi^{(v)}$ to zero, the optimal $\phi^{(v)}$ can be obtained by

$$\phi^{(v)}=\frac{2\,\mathrm{Tr}(S^TZ^{(v)})-\eta}{2\,\mathrm{Tr}\left((Z^{(v)})^TZ^{(v)}\right)}. \tag{31}$$

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Algorithm 1 Algorithm for Solving (11)

Input: Multi-view data matrices $\{X^{(v)} \in \mathbb{R}^{d_v \times n}\}_{v=1}^{m}$, Initial graphs $\{G^{(v)} \in \mathbb{R}^{n \times n}\}_{v=1}^{m}$, Number of clusters c, parameters α, β, γ

Output: The unified similarity matrix S

Initialize: $\phi^{(v)} = \frac{1}{m}$, $S = \sum_{v=1}^{m} \phi^{(v)} G^{(v)}$

1: **repeat**
2: Update F according to Eq. (12)
3: Update $Z^{(v)}$ by Eq. (17)
4: Update S according to Eq. (22)
5: Update $\phi^{(v)}$ according to Eq. (31)
6: **until** convergence

The Lagrange multiplier η can be determined by substituting the above expression into the constraint $\sum_{v=1}^{m} \phi^{(v)} = 1$.

Algorithm 1 summarizes the optimization procedure to solve (11).

V. Experiments

In this section, we perform comprehensive experiments to assess the performance of MCHL on various real-world benchmark datasets. In addition, we compare the proposed MCHL against several other state-of-the-art multi-view clustering (MVC) methods to validate its effectiveness.

A. Benchmark Datasets

In the experiments, six real-world multi-view datasets are used, which are listed below.

1) COIL20 [52] comprises 1440 grayscale images organized into 20 classes, with each class containing 72 images. Each image is represented by three feature sets, i.e., a 1024-D intensity feature set, a 3304-D LBP feature set, and a 6750-D Gabor feature set.
2) Caltech101-7 [53] is an image dataset designed for the object recognition task. Each image is associated with six feature views, including 48-D Gabor features, 40-D WM features, 254-D CENTRIST features, 1984-D HOG features, 512-D GIST features, and 928-D LBP features.
3) Caltech101-20 [53] is an image dataset used for object recognition, and it is a subset of Caltech101. The dataset provides a total of 2386 images divided into 20 categories. It is widely used to evaluate the performance of image classification and feature extraction algorithms.
4) UCI-3view [54] is obtained from the UCI machine learning repository. There are 2000 data samples in ten categories, and each sample is described by three different features.
5) NH-4660 [55] is a video face clustering dataset. It is derived from the movie Notting Hill. The dataset contains faces of five main cast members, including 4660 faces in 76 tracks.
6) Movies [56] is a dataset extracted from IMDb, which consists of 617 samples. Each sample is described by two different views (keywords and actors): one with a 1878-D feature set and the other with a 1398-D feature set.

B. Comparison Algorithms

We compare the proposed method with the following MVC algorithms in the experiments.

1) SFMC [TPAMI 2020] [23] proposes a scalable, parameter-free graph fusion framework that learns a joint graph and view weights automatically, using a connectivity constraint to directly identify clusters, achieving efficient and stable multi-view clustering without extensive hyperparameter tuning.
2) LMVSC [AAAI 2020] [2] learns small anchor graphs for each view and fuses them to perform spectral clustering on a reduced graph, enabling efficient and scalable multi-view subspace clustering with linear complexity.
3) MVSC [AAAI 2015] [57] uses bipartite graph approximation and local manifold fusion to integrate multi-view features efficiently, enabling large-scale spectral clustering with significant speedup over traditional methods.
4) DiMSC [CVPR 2015] [55] introduces a diversity term based on HSIC to enhance the complementarity among views, reducing redundancy and improving clustering performance through an efficient alternating optimization scheme.
5) SwMC [IJCAI 2017] [58] proposes a self-weighted MVC approach that learns view weights automatically and constructs a target graph with Laplacian rank constraint, enabling direct cluster assignment without postprocessing.
6) AMGL [IJCAI 2016] [59] automatically learns optimal weights for multiple graphs without introducing additional parameters, by reformulating the spectral learning model to better integrate heterogeneous views for multi-view clustering and semisupervised tasks.
7) FPMVS [TIP 2021] [34] unifies anchor selection and subspace graph construction into a single optimization framework, achieving efficient and parameter-free multi-view subspace clustering with linear time complexity, making it suitable for large-scale applications.
8) AWMVC [AAAI 2023] [60] learns dimension-flexible coefficient matrices from multi-view data and fuses them into an optimal consensus matrix via autoweighted optimization, enabling efficient large-scale clustering without hyperparameter tuning.
9) S2MVTC [CVPR 2024] [61] efficiently learns multi-view correlations via tensor feature fusion and a novel low-frequency approximation operator, enabling fast and accurate large-scale clustering.
10) DSTL [TMM 2025] [62] introduces a fast tensor-based method that disentangles semantic-related and unrelated features across views and aligns shared semantics to improve MVC performance without constructing graphs.
11) CTGL [TNNLS 2024] [63] learns collaborative topological graphs beyond conventional structures, thereby exploiting higher-order structural relationships across multiple views for clustering.
12) MAGIC [AAAI 2025] [64] defines anchors' diversity, balance, and compactness properties, designs Max-Mahalanobis Anchors via maximizing interanchor

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TABLE I

AVERAGE NMI OVER 20 RUNS BY DIFFERENT MVC ALGORITHMS. THE BEST SCORE ON EACH DATASET IS HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULT IS INDICATED WITH AN UNDERLINE

Method	*COIL20*	*Caltech101-7*	*Caltech101-20*	*UCI-3view*	*NH-p4660*	*Movies*
SFMC	$0.7924_{0.0000}$	$0.5378_{0.0000}$	$0.4698_{0.0000}$	$0.7520_{0.0000}$	$\underline{0.8411}_{0.0000}$	$0.2294_{0.0000}$
LMVSC	$0.7729_{0.0000}$	$0.4353_{0.0000}$	$0.5540_{0.0000}$	$0.7560_{0.0000}$	$0.6806_{0.0000}$	$0.2465_{0.0000}$
MVSC	$0.7568_{0.0433}$	$\underline{0.5446}_{0.1137}$	$0.5230_{0.0502}$	$0.6438_{0.0762}$	$0.5398_{0.1966}$	$0.2285_{0.0150}$
DiMSC	$0.8359_{0.0088}$	$0.5092_{0.0164}$	$0.5599_{0.0096}$	$0.7915_{0.0020}$	$0.7855_{0.0002}$	$0.3265_{0.0116}$
SwMC	$\underline{0.9429}_{0.0000}$	$0.5297_{0.0000}$	$0.4988_{0.0000}$	$0.6975_{0.0000}$	$0.0877_{0.0000}$	$0.1675_{0.0000}$
AMGL	$0.9124_{0.0273}$	$0.5056_{0.0841}$	$0.5356_{0.0328}$	$0.7778_{0.0396}$	$0.0819_{0.0282}$	$0.2272_{0.0088}$
FPMVS	$0.7463_{0.0000}$	$0.4732_{0.0000}$	$0.5849_{0.0000}$	$0.7349_{0.0000}$	$0.6715_{0.0000}$	$0.1451_{0.0000}$
AWMVC	$0.7694_{0.0000}$	$0.4064_{0.0000}$	$0.5551_{0.0000}$	$0.7331_{0.0000}$	$0.6261_{0.0000}$	$0.2488_{0.0000}$
S2MVTC	$0.8350_{0.0000}$	$0.5031_{0.0000}$	$\mathbf{0.7432}_{0.0000}$	$0.8587_{0.0000}$	$0.7354_{0.0000}$	$0.0721_{0.0000}$
DSTL	$0.8932_{0.0000}$	$0.7229_{0.0000}$	$\underline{0.6904}_{0.0000}$	$\underline{0.8969}_{0.0000}$	$0.7958_{0.0000}$	$\underline{0.3573}_{0.0000}$
CTGL	$0.9189_{0.0000}$	$0.5474_{0.0000}$	$0.5967_{0.0000}$	$0.7714_{0.0000}$	$0.2523_{0.0000}$	$0.2918_{0.0000}$
MAGIC	$0.7749_{0.0000}$	$0.5681_{0.0000}$	$0.3303_{0.0000}$	$0.5390_{0.0000}$	$0.6556_{0.0000}$	$0.2167_{0.0000}$
SLWSE-AGL	$0.7599_{0.0000}$	$0.6495_{0.0000}$	$0.6323_{0.0000}$	$0.8648_{0.0000}$	$0.7900_{0.0000}$	$0.1530_{0.0000}$
Ours	$\mathbf{0.9429}_{0.0000}$	$\mathbf{0.7575}_{0.0000}$	$0.6303_{0.0000}$	$\mathbf{0.9321}_{0.0000}$	$\mathbf{0.8525}_{0.0000}$	$\mathbf{0.3686}_{0.0000}$

TABLE II

AVERAGE ARI OVER 20 RUNS BY DIFFERENT MVC ALGORITHMS. THE BEST SCORE ON EACH DATASET IS HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULT IS INDICATED WITH AN UNDERLINE

Method	*COIL20*	*Caltech101-7*	*Caltech101-20*	*UCI-3view*	*NH-p4660*	*Movies*
SFMC	$0.5445_{0.0000}$	$0.4230_{0.0000}$	$0.3246_{0.0000}$	$0.6705_{0.0000}$	$\underline{0.8366}_{0.0000}$	$0.0011_{0.0000}$
LMVSC	$0.5796_{0.0000}$	$0.3591_{0.0000}$	$0.4252_{0.0000}$	$0.6675_{0.0000}$	$0.6211_{0.0000}$	$0.0833_{0.0000}$
MVSC	$0.5195_{0.0871}$	$0.4792_{0.1414}$	$0.3528_{0.1126}$	$0.4958_{0.1114}$	$0.4817_{0.2318}$	$0.0580_{0.0123}$
DiMSC	$0.7130_{0.0184}$	$0.4313_{0.0218}$	$0.3281_{0.0171}$	$0.7490_{0.0038}$	$0.7883_{0.0003}$	$\mathbf{0.1684}_{0.0150}$
SwMC	$\underline{0.8353}_{0.0000}$	$0.3858_{0.0000}$	$0.2492_{0.0000}$	$0.5063_{0.0000}$	$0.0288_{0.0000}$	$0.0186_{0.0000}$
AMGL	$0.7486_{0.0785}$	$0.4461_{0.1240}$	$0.3295_{0.0432}$	$0.6776_{0.0756}$	$0.0276_{0.0215}$	$0.0589_{0.0120}$
FPMVS	$0.5321_{0.0000}$	$0.4134_{0.0000}$	$0.4087_{0.0000}$	$0.6458_{0.0000}$	$0.6093_{0.0000}$	$0.0219_{0.0000}$
AWMVC	$0.6265_{0.0000}$	$0.3116_{0.0000}$	$0.3171_{0.0000}$	$0.6647_{0.0000}$	$0.5849_{0.0000}$	$0.0890_{0.0000}$
S2MVTC	$0.6603_{0.0000}$	$0.3348_{0.0000}$	$\underline{0.5128}_{0.0000}$	$0.7697_{0.0000}$	$0.5977_{0.0000}$	$0.0236_{0.0000}$
DSTL	$0.7941_{0.0000}$	$\underline{0.8540}_{0.0000}$	$0.3830_{0.0000}$	$\underline{0.8588}_{0.0000}$	$0.6750_{0.0000}$	$\underline{0.1557}_{0.0000}$
CTGL	$0.7796_{0.0000}$	$0.5100_{0.0000}$	$0.4061_{0.0000}$	$0.6623_{0.0000}$	$0.1206_{0.0000}$	$0.1111_{0.0000}$
MAGIC	$0.5940_{0.0000}$	$0.5748_{0.0000}$	$0.3303_{0.0000}$	$0.5390_{0.0000}$	$0.6556_{0.0000}$	$0.0374_{0.0000}$
SLWSE-AGL	$0.5660_{0.0000}$	$0.4889_{0.0000}$	$\mathbf{0.5900}_{0.0000}$	$0.8338_{0.0000}$	$0.8226_{0.0000}$	$0.0281_{0.0000}$
Ours	$\mathbf{0.8512}_{0.0000}$	$\mathbf{0.8937}_{0.0000}$	$0.4939_{0.0000}$	$\mathbf{0.9302}_{0.0000}$	$\mathbf{0.8658}_{0.0000}$	$0.0195_{0.0000}$

TABLE III

AVERAGE ACC OVER 20 RUNS BY DIFFERENT MVC ALGORITHMS. THE BEST SCORE ON EACH DATASET IS HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULT IS INDICATED WITH AN UNDERLINE

Method	*COIL20*	*Caltech101-7*	*Caltech101-20*	*UCI-3view*	*NH-p4660*	*Movies*
SFMC	$0.6438_{0.0000}$	$0.6398_{0.0000}$	$0.5750_{0.0000}$	$0.7520_{0.0000}$	$\mathbf{0.9189}_{0.0000}$	$0.0891_{0.0000}$
LMVSC	$0.6549_{0.0000}$	$0.4505_{0.0000}$	$0.5067_{0.0000}$	$0.8030_{0.0000}$	$0.7376_{0.0000}$	$0.2674_{0.0000}$
MVSC	$0.6166_{0.0658}$	$0.6368_{0.0614}$	$0.5123_{0.0624}$	$0.6195_{0.0811}$	$0.6293_{0.1453}$	$0.2519_{0.0131}$
DiMSC	$0.7631_{0.0173}$	$0.5714_{0.0238}$	$0.4530_{0.0245}$	$0.8651_{0.0037}$	$0.8427_{0.0002}$	$\mathbf{0.3354}_{0.0172}$
SwMC	$0.8639_{0.0000}$	$0.6479_{0.0000}$	$0.5721_{0.0000}$	$0.6550_{0.0000}$	$0.3388_{0.0000}$	$0.2091_{0.0000}$
AMGL	$0.8109_{0.0416}$	$0.7072_{0.0697}$	$0.5519_{0.0239}$	$0.7575_{0.0765}$	$0.3629_{0.0184}$	$0.2617_{0.0139}$
FPMVS	$0.6375_{0.0000}$	$0.5455_{0.0000}$	$0.5394_{0.0000}$	$0.7220_{0.0000}$	$0.7131_{0.0000}$	$0.1702_{0.0000}$
AWMVC	$0.7076_{0.0000}$	$0.4098_{0.0000}$	$0.4392_{0.0000}$	$0.7775_{0.0000}$	$0.7408_{0.0000}$	$0.2512_{0.0000}$
S2MVTC	$0.6965_{0.0000}$	$0.4172_{0.0000}$	$0.5876_{0.0000}$	$0.7740_{0.0000}$	$0.6966_{0.0000}$	$0.1524_{0.0000}$
DSTL	$\underline{0.8806}_{0.0000}$	$\underline{0.8229}_{0.0000}$	$0.5029_{0.0000}$	$\underline{0.9305}_{0.0000}$	$0.7322_{0.0000}$	$\underline{0.3193}_{0.0000}$
CTGL	$0.7826_{0.0000}$	$0.6486_{0.0000}$	$0.5453_{0.0000}$	$0.7420_{0.0000}$	$0.4461_{0.0000}$	$0.2934_{0.0000}$
MAGIC	$0.6979_{0.0000}$	$0.6594_{0.0000}$	$0.4426_{0.0000}$	$0.6725_{0.0000}$	$0.7957_{0.0000}$	$0.2318_{0.0000}$
SLWSE-AGL	$0.6250_{0.0000}$	$0.6045_{0.0000}$	$\underline{0.6433}_{0.0000}$	$0.9180_{0.0000}$	$\underline{0.8957}_{0.0000}$	$0.1767_{0.0000}$
Ours	$\mathbf{0.8917}_{0.0000}$	$\mathbf{0.9342}_{0.0000}$	$\mathbf{0.6693}_{0.0000}$	$\mathbf{0.9685}_{0.0000}$	$0.8597_{0.0000}$	$0.0794_{0.0000}$

Mahalanobis distance, and guides cross-view representations to align with such anchors for accurate large-scale MVC.

13) SLWSE-AGL [TCSVT 2026] [65] incorporates sample-level weighting and structure-enhanced anchor graph learning to improve the robustness and scalability of multi-view clustering.

C. Evaluation Metrics

In this work, we use four widely adopted evaluation metrics to measure the effectiveness of our method, including normalized mutual information (NMI), adjusted Rand index (ARI), clustering accuracy (ACC), and purity (PUR).

D. Clustering Results

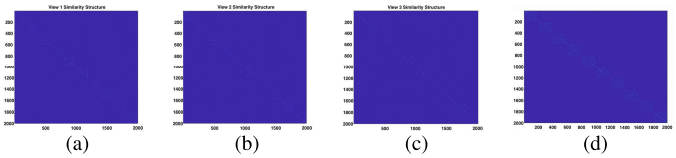
Before the experiments, we construct the initial graphs $G^{(v)}$ by selecting the five-nearest neighbors from the original data matrices $X^{(v)}$. The performance of different methods in terms of NMI, ARI, ACC, and PUR, on different datasets, is presented in Tables I–IV, respectively. Average metrics and their standard deviations are reported for different methods

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TABLE IV

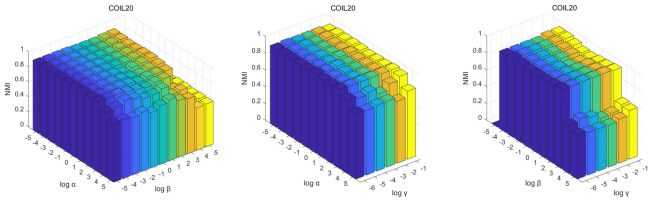
AVERAGE PUR OVER 20 RUNS BY DIFFERENT MVC ALGORITHMS. THE BEST SCORE ON EACH DATASET IS HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULT IS INDICATED WITH AN UNDERLINE

Method	*COIL20*	*Caltech101-7*	*Caltech101-20*	*UCI-3view*	*NH-p4660*	*Movies*
SFMC	$0.7021_{0.0000}$	$0.8541_{0.0000}$	$0.6391_{0.0000}$	$0.7530_{0.0000}$	$\mathbf{0.9189}_{0.0000}$	$0.2901_{0.0000}$
LMVSC	$0.6813_{0.0000}$	$0.8616_{0.0000}$	$0.7670_{0.0000}$	$0.8030_{0.0000}$	$0.8120_{0.0000}$	$0.2836_{0.0000}$
MVSC	$0.6469_{0.0564}$	$0.8291_{0.0785}$	$0.6700_{0.0476}$	$0.6315_{0.0848}$	$0.6524_{0.1391}$	$0.2811_{0.0137}$
DiMSC	$0.7914_{0.0102}$	$0.8843_{0.0053}$	$0.7873_{0.0089}$	$\underline{0.8651}_{0.0037}$	$0.8505_{0.0001}$	$\underline{0.3685}_{0.0121}$
SwMC	$\underline{0.8986}_{0.0000}$	$0.8365_{0.0000}$	$0.6987_{0.0000}$	$0.6905_{0.0000}$	$0.3710_{0.0000}$	$0.2334_{0.0000}$
AMGL	$0.8597_{0.0343}$	$0.8324_{0.0641}$	$0.6836_{0.0235}$	$0.7722_{0.0574}$	$0.3707_{0.0210}$	$0.2850_{0.0119}$
FPMVS	$0.6542_{0.0000}$	$0.8474_{0.0000}$	$0.7498_{0.0000}$	$0.7345_{0.0000}$	$0.7850_{0.0000}$	$0.1864_{0.0000}$
AWMVC	$0.7264_{0.0000}$	$0.8182_{0.0000}$	$0.7842_{0.0000}$	$0.7905_{0.0000}$	$0.7693_{0.0000}$	$0.2853_{0.0000}$
S2MVTC	$0.7431_{0.0000}$	$0.8799_{0.0000}$	$0.7980_{0.0000}$	$0.8000_{0.0000}$	$0.8049_{0.0000}$	$0.1524_{0.0000}$
DSTL	$0.8806_{0.0000}$	$0.8799_{0.0000}$	$\mathbf{0.8282}_{0.0000}$	$\underline{0.9305}_{0.0000}$	$0.8624_{0.0000}$	$0.3452_{0.0000}$
CTGL	$0.8479_{0.0000}$	$0.8562_{0.0000}$	$0.7821_{0.0000}$	$0.7580_{0.0000}$	$0.4861_{0.0000}$	$0.3387_{0.0000}$
MAGIC	$0.7063_{0.0000}$	$0.8630_{0.0000}$	$0.7544_{0.0000}$	$0.6855_{0.0000}$	$0.7957_{0.0000}$	$0.2642_{0.0000}$
SLWSE-AGL	$0.6458_{0.0000}$	$\underline{0.9064}_{0.0000}$	$0.7619_{0.0000}$	$0.9180_{0.0000}$	$0.8618_{0.0000}$	$0.1994_{0.0000}$
Ours	$\mathbf{0.8986}_{0.0000}$	$\mathbf{0.9342}_{0.0000}$	$\underline{0.7984}_{0.0000}$	$\mathbf{0.9685}_{0.0000}$	$\underline{0.9090}_{0.0000}$	$\mathbf{0.5348}_{0.0000}$



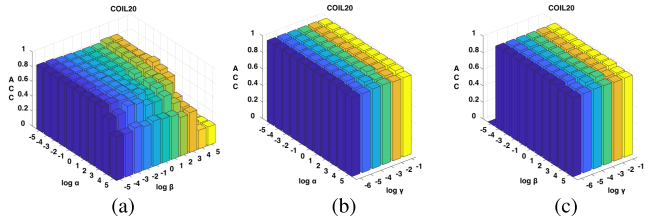
(a) (b) (c) (d)

Fig. 2. Similarity matrices from multiple views and the unified MCHL representation. (a) View 1. (b) View 2. (c) View 3. (d) Unified.



(a) (b) (c)

Fig. 3. NMI of MCHL with different α, β, and γ on the COIL20 dataset. (a) $\gamma = 1$. (b) $\alpha = 1$. (c) $\beta = 1$.



(a) (b) (c)

Fig. 4. ACC of MCHL with different α, β, and γ on the COIL20 dataset. (a) $\gamma = 1$. (b) $\alpha = 1$. (c) $\beta = 1$.

across various datasets. The results show that MCHL achieves the best performance in most cases. This demonstrates the effectiveness of jointly leveraging first-order and topological structures to identify the underlying cluster structures.

The superior performance of the proposed method can be attributed to its hybrid-order similarity learning strategy. Most existing graph-based clustering methods mainly rely on first-order neighborhood relationships derived from pairwise similarities, which are effective in preserving local geometric structures but are often sensitive to noise and insufficient to characterize complex sample relationships. By contrast, the proposed method simultaneously incorporates first-order neighborhood information and second-order topological correlations induced by shared neighbors. This joint modeling strategy enables the learned similarity graph to preserve local consistency while capturing higher-order structural dependencies, resulting in a more robust and discriminative representation for clustering.

Moreover, the integration of first-order and topological structures is particularly beneficial for multi-view data with heterogeneous feature distributions. In such scenarios, relying solely on first-order similarities may lead to unstable graph constructions due to view-specific noise or feature distortions. The topological structure serves as a complementary constraint that enhances structural consistency across views, which probably leads to the consistent performance gains observed on several challenging datasets.

Fig. 2 visualizes the consensus graph S learned on the UCI-3view dataset. Based on this consensus graph, the cluster label for each data point can be explicitly observed. Thus Fig. 2 can be interpreted as a visualization of the connectivity effect.

Different from conventional approaches that perform spectral clustering as a postprocessing step, the proposed method explicitly enforces the connectivity constraint during the consensus graph learning stage. By imposing the Laplacian rank constraint, the learned graph is encouraged to form a predefined number of connected components, which directly correspond to clustering groups. This design leads to a graph representation that is inherently clustering-friendly and reduces the ambiguity introduced by subsequent spectral decomposition.

From this visualization, we can see that our method produces a block diagonal structure and correctly identifies the number of diagonal blocks. This demonstrates that the method can yield a consensus graph with a clear clustering structure, leading to better clustering results.

E. Parameter Sensitivity

This section investigates the performance of the proposed method by studying its sensitivity with respect to different

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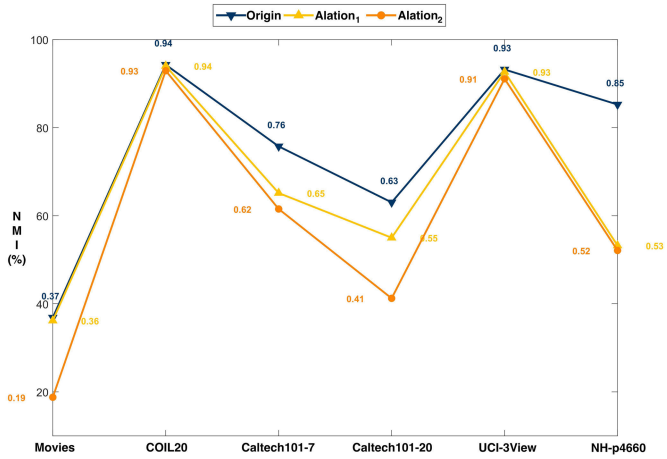


Fig. 5. Ablation study of the proposed method on six benchmark multi-view datasets.

parameters. There are three parameters in our model as formulated in (11) (i.e., α, β, and γ). We do not need to search for the parameter λ, as it can be tuned heuristically. When λ is initialized to a random positive value, our model is capable of automatically halving or doubling it in each iteration when the number of connected components in S is greater than or less than the number of clusters c. We fix one parameter at a time and then vary the remaining parameters to explore the performance of the method. The experimental results shown in Figs. 3 and 4 demonstrate that MCHL is relatively insensitive to these three hyperparameters.

F. Ablation Studies

The similarity learning of this method consists of two parts: first-order neighborhood structure learning and topological neighborhood structure learning. To evaluate the contribution of each structural component, we construct two ablation variants of the proposed model, denoted as $Ablation_1$ and $Ablation_2$. In particular, $Ablation_1$ removes the first-order neighborhood structure learning module, while $Ablation_2$ removes the topological neighborhood structure learning module. Fig. 5 reports the clustering performance measured by the NMI metric across six benchmark datasets. The results indicate that when key components of the method are ablated, the clustering performance declines, demonstrating the effectiveness of both the first-order and topological structure learning components.

G. Convergence Analysis

To evaluate the stability and reliability of the proposed algorithm, we conducted convergence experiments on various datasets. In particular, we monitored the PUR values over successive iterations. As shown in Fig. 6, the optimization process consistently converges within a reasonable number of iterations, indicating the robustness of the model. These results demonstrate the effectiveness of the optimization strategy and validate the empirical convergence properties of the proposed method.

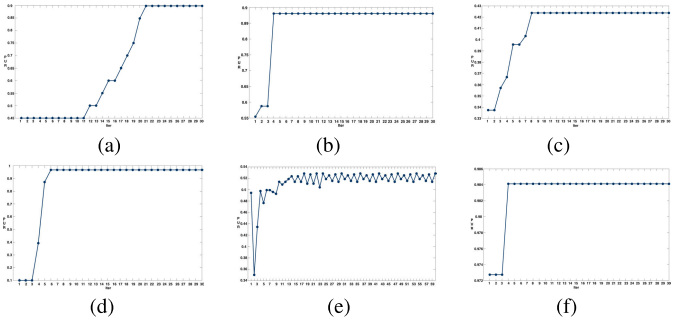


Fig. 6. Convergence curves of the proposed method in terms of PUR. (a) COIL20. (b) Caltech101-7. (c) Caltech101-20. (d) UCI-3view. (e) Movies. (f) NH-4660.

VI. Conclusion

In this article, we propose a novel MVC method called MCHL, which jointly utilizes information from both first-order neighborhood information and second-order topological information to capture a more comprehensive cluster structure. MCHL integrates view-specific graph learning, consensus graph learning, and spectral clustering within a unified framework. In this framework, the model jointly learns first-order and topological similarities for each view and fuses the resulting view-specific graphs into a consensus graph. Based on this unified graph, we incorporate the spectral clustering task by imposing a rank constraint on the Laplacian matrix, thereby ensuring that the number of connected components in the unified graph corresponds to the number of clusters. Extensive experiments on multiple benchmark datasets demonstrate that MCHL consistently outperforms the baseline algorithms across several evaluation metrics.

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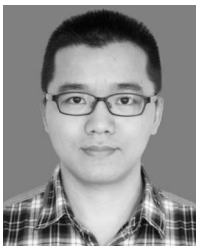
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